

ORIGINAL CONTRIBUTION

AI-Derived LA Volume Index, LA/RA and LA/LV Volume Ratios From Coronary Artery Calcium Scans Predict Long-Term Atrial Fibrillation and Stroke

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BACKGROUND: The AI-CVD initiative aims to extract actionable insights from coronary artery calcium (CAC) scans beyond the traditional CAC score. Although AI-derived cardiac chamber volumes predict atrial fibrillation (AF) and stroke, the long-term prognostic value of chamber ratios is less established. We evaluated the predictive value of AI-derived left atrial volume index and related chamber ratios (left atrial [LA]/right atrial [RA], LA/left ventricular) from routine CAC scans for incident AF and stroke, and their incremental value beyond established risk scores.

METHODS: Pooled participant-level data from 2 prospective cohorts, the MESA (Multi-Ethnic Study of Atherosclerosis, 2000–2002, n=5670) and the FHS (Framingham Heart Study Offspring cohort, 1998–2001, n=1142), were analyzed. Primary outcomes were incident AF and incident stroke. AI-enabled volumetry (AutoChamber, AI-CVD platform) quantified cardiac chamber metrics from noncontrast CAC scans. Cox proportional hazards models, net reclassification improvement, time-dependent area under the curve, calibration metrics, and least absolute shrinkage and selection operator regression were applied to evaluate predictive performance.

RESULTS: Over a median 17-year follow-up, 1302 participants developed AF, and 365 experienced stroke events. Individuals in the ≥95th percentile of chamber metrics had a significantly increased risk. Adjusted hazard ratios for AF were 2.66 (95% CI, 2.23–3.17) for left atrial volume index, 2.04 (95% CI, 1.71–2.45) for LA/left ventricular (LV) ratio, and 1.87 (95% CI, 1.55–2.26) for LA/RA ratio. For stroke, corresponding hazard ratios were 1.96 (95% CI, 1.38–2.77), 1.64 (95% CI, 1.15–2.33), and 1.83 (95% CI, 1.29–2.59), respectively. AI-derived metrics improved reclassification beyond Cohorts for Heart and Aging Research in Genomic Epidemiology Atrial Fibrillation risk score and Framingham Stroke Risk Profile, with greatest improvements for AF from left atrial volume index (net reclassification improvement, 0.48) and stroke from LA/RA ratio (net reclassification improvement, 0.39), driven mainly by nonevent classification. Although discrimination improvements (area under the curve) were modest, chamber measurements substantially improved Framingham Stroke Risk Profile calibration (slope, 0.448 to 0.834–0.902). Among all chamber metrics (including volumes and ratios), the least absolute shrinkage and selection operator identified left atrial volume index as the strongest predictor for AF, and LA/RA ratio as the strongest for stroke.

CONCLUSIONS: AI-enabled left atrial volumetric and ratio-based metrics derived opportunistically from CAC scans provide incremental predictive value for AF and stroke prediction.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: atrial fibrillation ■ artificial intelligence ■ calibration ■ global health ■ prevalence

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Nonstandard Abbreviations and Acronyms

AF	atrial fibrillation
AI	artificial intelligence
AUC	area under the receiver operating characteristic curve
BSA	body surface area
CAC	coronary artery calcium
CHARGE-AF	Cohorts for Heart and Aging Research in Genomic Epidemiology Atrial Fibrillation risk score
CHD	coronary heart disease
CMR	cardiac magnetic resonance
CT	computed tomography
EBCT	electron-beam computed tomography
FHS	Framingham Heart Study Offspring cohort
FSRP	Framingham Stroke Risk Profile
IBS	integrated Brier score
LA	left atrial/left atrium
LASSO	least absolute shrinkage and selection operator
LAVI	left atrial volume index
LV	left ventricle/left ventricular
MDCT	multidetector computed tomography
MESA	Multi-Ethnic Study of Atherosclerosis
NRI	net reclassification improvement
NT-proBNP	N-terminal pro-B-type natriuretic peptide N-terminal pro-brain natriuretic peptide
RA	right atrium/right atrial

Atrial fibrillation (AF) and stroke represent significant global health burdens that continue to escalate worldwide. The 2019 global burden of AF is estimated at 59.7 million cases, representing a doubling since 1990, with projections indicating the prevalence will reach 15.9 million in America by 2050 and 17.9 million in Europe by 2060.^{1,2} Stroke, as the second leading cause of death globally, affects ≈12.2 million people annually, with 6.6 million deaths and 143 million disability-adjusted life-years lost in 2019.^{3,4} The growing incidence of both conditions, particularly in aging populations and those with increasing cardiovascular risk factors, underscores the urgent need for improved prediction and prevention strategies.

The left atrial (LA) size has emerged as a predictor of both AF and stroke risk across multiple imaging modalities. Meta-analyses have consistently demonstrated that LA enlargement is associated with increased stroke risk, with each 1 cm increase in LA diameter associated with a 24% increase in stroke odds.⁵ In cardiac magnetic

resonance (CMR) imaging studies, elevated LA maximum volume index has been reported to be associated with incident AF, with hazard ratios of 1.38 per SD.⁶ Although CMR is considered the gold standard for assessing chamber volumes and ventricular geometry, availability is limited, and high cost compromises feasibility for widespread use in prevention and screening settings.⁷ On the other hand, the number of coronary artery calcium (CAC) scans is steadily increasing,^{8,9} which may allow opportunistic evaluation of cardiac chamber size.

We previously demonstrated the potential of artificial intelligence (AI) applied to CAC scans for comprehensive cardiovascular risk assessment. We have developed the AI-CVD platform, which includes automated cardiac chamber volumetry (AutoChamber) that can extract valuable information from routine CAC scans beyond the traditional Agatston score. In prior studies in which AI was applied to the MESA (Multi-Ethnic Study of Atherosclerosis) cohort, we reported that AI-enabled LA volumetry predicted AF and stroke as well as LA volume measured by human experts using CMR imaging, with both approaches outperforming Cohorts for Heart and Aging Research in Genomic Epidemiology Atrial Fibrillation (CHARGE-AF), NT-proBNP (N-terminal pro-B-type natriuretic peptide), and the Agatston score.^{10,11} We have also shown that AI-enabled cardiac chamber volumetry can predict heart failure, outperforming NT-proBNP.^{12,13} Additionally, these volumetry findings were associated with the prediction of risk of future coronary heart disease above and beyond the Agatston score.⁸

Although LA size is an established predictor, emerging evidence suggests that LA-related chamber ratios may provide incremental predictive value for cardiovascular outcomes.^{14–16} However, the predictive utility of AI-derived LA/right atrial (RA) and LA/LV ratios from CAC scans for AF and stroke prediction remains largely unexplored. Chamber ratios may offer theoretical advantages by reflecting more precise pathophysiological relationships, potentially capturing subtle alterations in atrial-ventricular coupling and inter-atrial remodeling that isolated chamber volumes may not detect. In this study, we evaluate the value of LA chamber and ratio measures derived from AI-enabled CAC scan volumetry across both MESA and FHS (Framingham Heart Study) offspring cohorts for the prediction of incident AF and stroke. By leveraging these 2 diverse and well-characterized cohort studies, we aim to establish robust, generalizable findings that could inform clinical practice and enhance cardiovascular risk stratification strategies.

METHODS

Availability of Data and Materials

MESA and FHS data are available on approval of an article proposal and completion of a Data and Materials Transfer agreement.

Study Population

Individual participant-level data were pooled from 2 prospective cohort studies: the MESA and the FHS Offspring cohort. Study designs, participant characteristics, and event adjudication procedures have been previously reported.^{17,18}

Briefly, MESA is a multiethnic, community-based cohort study that enrolled 6814 participants aged 45 years–84 years with no known cardiovascular disease. Initial enrollment occurred from 2000 to 2002 across 6 US field centers: Baltimore, MD; Los Angeles, CA; Chicago, IL; Forsyth County, NC; New York City, NY; and St. Paul, MN. Participants underwent comprehensive baseline assessments and ECG-gated computed tomography (CT) scans to determine CAC scores.

The FHS Offspring cohort, initiated in 1971, enrolled 5124 individuals with follow-up assessments conducted approximately every 4 years. CAC assessment using multidetector CT (MDCT) was first incorporated during the seventh examination cycle (1998–2001) among 1418 FHS Offspring cohort participants.

Participants were excluded from both cohorts if they had a history of AF or prior stroke at baseline, declined authorization for commercial data use, had incomplete CT scans, or had missing data on chamber volume measurements, missing cardiovascular outcome status, or time-to-event information (Figure S1).

All participants in both cohorts provided written informed consent at recruitment. Ethical approval was obtained from Institutional Review Boards at all participating centers, and the study adhered to the ethical principles outlined in the Declaration of Helsinki. This analysis follows the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines.

CT Scan Protocol

In MESA, baseline noncontrast cardiac CT scans were performed using 2 different modalities across the 6 field centers. Electron-beam CT (EBCT) scanners were employed in New York, Illinois, and California, while MDCT systems were used in Maryland, Minnesota, and North Carolina. EBCT scans were prospectively gated at 50% of the R-R interval, whereas MDCT scans used 80% gating. All scans were acquired during a mid-inspiratory breath-hold and extended from the carina through the apex of the heart, capturing adjacent thoracic structures including the liver and spleen. Reconstruction parameters were standardized across all sites: 2.5 mm slice thickness, 512×512 matrix, 350 mm field of view, and 0.68 mm in-plane pixel spacing.¹⁹ In the FHS Offspring cohort, CAC imaging was performed using an 8-slice MDCT scanner (LightSpeed Ultra, GE Healthcare, Milwaukee, WI) with prospective ECG-triggering at 50% of the R-R interval during a practiced mid-inspiratory breath-hold (≈18 seconds). Each scan included 48 contiguous 2.5-mm slices acquired at 120 kVp, with tube current adjusted by body weight (320 mA for <220 lbs and 400 mA for ≥220 lbs), and a gantry rotation time of 500 ms. The average effective radiation dose was 1.0–1.25 mSv. Images were reconstructed using a 35 cm field of view.²⁰

AI-Enabled Chamber Volumetry Using CAC Scans

AutoChamber (Version 2.0; HeartLung.AI, Houston, TX) is based on the high-resolution TotalSegmentator cardiac

chambers model, utilizing the nnU-Net framework for automated segmentation of major cardiac structures, including the myocardium, left and right atria, left and right ventricles, aorta, and pulmonary artery.²¹ The model received FDA Breakthrough designation for automated cardiac chamber volume measurement in noncontrast chest CT scans and enables fully automated segmentation and volumetric assessment from standard noncontrast ECG-gated CAC scans, without the need for intravenous contrast or dedicated cardiac imaging protocols (Figure 1).

The base model was trained on 1228 contrast-enhanced chest and thoracic CT scans from University Hospital Basel, encompassing a wide variety of scanner types, imaging protocols, and acquisition parameters to enhance robustness and generalizability. All training images were acquired at sub-millimeter resolution and manually annotated by expert raters. The data set was partitioned into 90% training, 5% validation, and 5% independent testing sets. For AutoChamber development, iterative training was employed using matched noncontrast and contrast-enhanced ECG-gated cardiac CT images (slice thickness 1.5 mm) from the same individuals acquired within the same session. Human supervisors corrected model errors, and revised annotations were used to progressively enhance model accuracy.

Before training, CT volumes were resampled to isotropic voxel spacing, cropped to the heart region with a 20 mm padding margin, and preprocessed using intensity normalization and foreground cropping. The model was trained using a composite Dice and Cross-Entropy loss function, with data augmentation strategies including random spatial transformations to improve generalizability.

Quality control procedures were applied to identify and flag potentially invalid segmentations. The module detects instances where the segmentation mask intersects with the top or bottom edge of the CT volume, which may indicate incomplete capture of cardiac structures.

AutoChamber has undergone extensive multimodal validation. Reproducibility was evaluated using duplicate ECG-gated noncontrast CT scans from MESA participants, acquired 2 minutes apart, demonstrating excellent inter-scan agreement across different scanner types (Figure S2). The model's accuracy has been validated against CMR imaging and contrast-enhanced CT scans.^{10,12}

The LA volume index (LAVI) was calculated by indexing the LA volume to BSA, while the LA/LV and LA/RA volume ratios were computed by dividing the LA volume by LV and RA volumes, respectively. These chamber-specific indices were assessed as continuous and categorical variables and were used to explore their associations with incident AF and stroke.

Outcome Definition

Incident AF was systematically identified through comprehensive surveillance in both cohorts. In MESA, participants were contacted every 9 to 12 months to report new cardiovascular diagnoses. Incident AF was identified using International Classification of Diseases codes 427.3x (*International Classification of Diseases* [ICD-9]) or I48.x (ICD-10) from inpatient stays and Medicare claims. For fee-for-service Medicare participants, AF was defined by codes 427.31 or 427.32 in any position, excluding AF hospitalizations associated with open cardiac surgery.¹⁸ In FHS, AF was diagnosed when AF or flutter

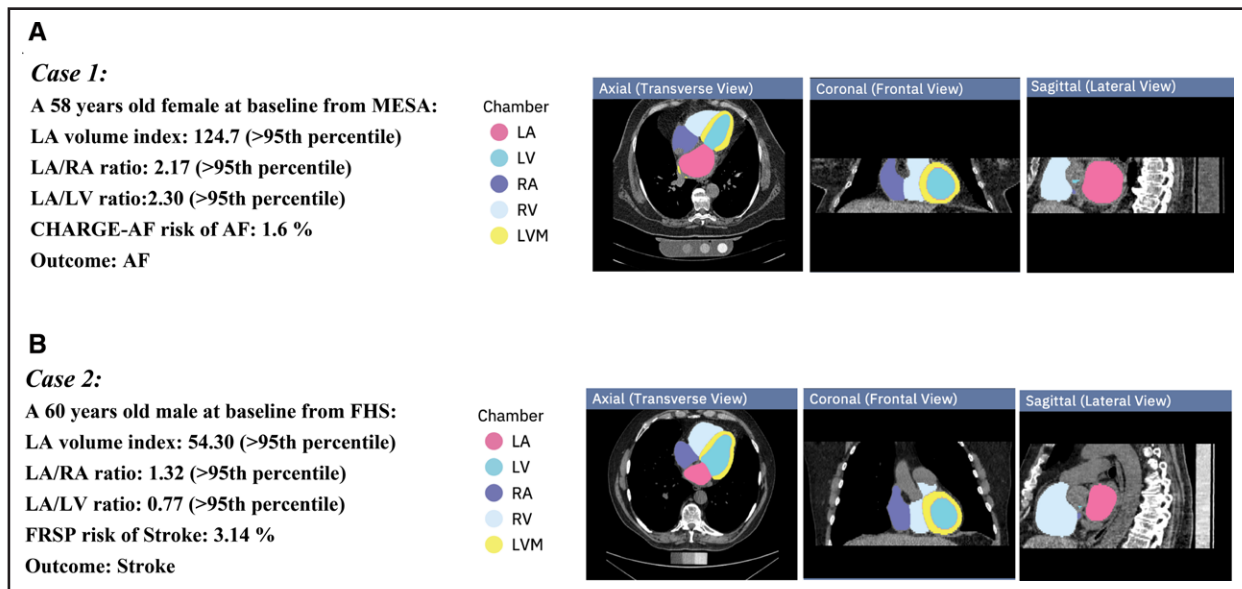


Figure 1. Coronary artery calcium (CAC) scans with artificial intelligence (AI)-derived left atrial (LA) chamber metrics and ratios. The AI segmentation of noncontrast CAC images yields the left atrial volume index (LAVI) and atrial ratios (LA/right atrial [RA] and LA/left ventricle [LV]). **A**, Case of incident atrial fibrillation; and **B** shows a case of incident stroke.

were present on electrocardiograms from clinic visits, outpatient visits, hospitalizations, or Holter monitors. All potential cases were adjudicated by FHS cardiologists using medical records from the research center, hospitals, and clinics.²² At the time of analysis, median follow-up was 16.5 years (interquartile range [IQR], 10.8–17.4; maximum, 18.5) for MESA and 14.2 years (IQR, 12.4–15.6; maximum, 17.3) for FHS.

Incident stroke events included both ischemic and hemorrhagic strokes, rigorously adjudicated by trained physicians and neurologists. Stroke was defined as a rapid onset of documented focal neurological deficit lasting ≥ 24 hours or until death, or if < 24 hours, the presence of a clinically relevant brain imaging lesion. Patients with focal deficits secondary to trauma, tumor, infection, or other nonvascular causes were excluded. In MESA, interim hospitalizations were assessed every 9–12 months, with stroke events adjudicated by committee review using medical records and death certificates, and evaluation by 3 vascular neurologists.¹⁸ In FHS, study visits occurred approximately every 4-to-6 year with detailed medical history collection. Concerning events were reviewed by stroke-expert neurologists, followed by in-person evaluation when indicated, and panel review by at least 2 stroke neurologists.²³ At the time of analysis, median follow-up was 17.7 years (IQR, 12.3–18.5; maximum, 19.4) for MESA and 14.4 years (IQR, 12.9–15.6; maximum, 17.3) for FHS.

Statistical Analysis

Analyses were performed using R Studio (version 4.3.0) and Python 3.10. Descriptive statistics were presented as mean $\pm$ SD for normally distributed continuous variables, median (25th–75th percentile) for non-normally distributed variables, and frequencies (percentages) for categorical variables.

We used complete case analysis without multiple imputation. Participants with missing predictor or outcome data were excluded. In adjusted models, cases with missing covariates were excluded from those specific analyses.

The correlations between AI-derived chamber metrics were analyzed using Pearson correlation analysis and illustrated using a symmetrical correlation matrix (Figure S3).

Restricted cubic spline analyses with 3 knots at the 25th, 50th, and 75th percentiles examined potential nonlinear associations between AI-derived chamber metrics and outcomes using age- and sex-adjusted Cox proportional hazards models.

LA-related metrics (LAVI, LA/RA ratio, LA/LV ratio) were categorized using sex-specific thresholds due to significant between-sex differences. Three categories were defined: 25th to 75th percentiles (reference), >75th percentile, and >95th percentile (Table S1). The IQR served as a reference based on S-shaped and J-shaped relationships observed in preliminary restricted cubic spline analyses (Figures S4 and S5). Furthermore, the distribution of each >95th percentile of LA-related metrics was illustrated using a Venn diagram for AF and stroke (Figure S6).

Cox proportional hazards regression assessed associations between categorized LA-related chamber metrics and incident AF and stroke. Penalized Cox regression with L2 regularization enhanced model stability and addressed potential collinearity. Model assumptions were validated using Schoenfeld residuals for proportional hazards and martingale residuals for linearity. Hazard ratios per SD increment were calculated for continuous relationships. Fully adjusted models included age, sex, race and ethnicity, systolic and diastolic blood pressure, antihypertensive medication use, low-density lipoprotein cholesterol, high-density lipoprotein cholesterol, triglycerides, lipid-lowering medication use, diabetes, and current smoking status. In the pooled cohorts, we also adjusted for the cohort factor (MESA versus FHS).

Prespecified subgroup analyses were performed by stratifying participants according to age (≤ 60 versus > 60 years), sex, race and ethnicity, and cardiovascular risk factors (current smoking, hypertension, dyslipidemia, diabetes) to evaluate the consistency of associations across clinically relevant population strata.

The added predictive utility of AI-measured chamber metrics (LAVI, LA/RA ratio, LA/LV ratio) was evaluated by comparing base models (CHARGE-AF for 5-year AF prediction,²⁴ Framingham stroke risk profile [FSRP] for 10-year stroke prediction²⁵) to models incorporating these metrics as continuous variables or dichotomized at >75th and >95th percentiles. Category-free net reclassification improvement (NRI) was calculated at 5 years for AF and 10 years for stroke as the sum of differences between upward and downward reclassifications for events and nonevents. Discrimination improvement was assessed using time-dependent area under the receiver operating characteristic curve (AUC) at the same time points. Model calibration was assessed by comparing predicted versus observed event probabilities using calibration slope and integrated Brier scores, where a calibration slope of 1.0 indicates perfect calibration (values >1.0 suggest underestimation in high-risk groups; values <1.0 suggest overestimation) and lower integrated Brier score values indicate better overall prediction accuracy. All analyses were performed separately for MESA, FHS, and pooled cohorts.

Cumulative incidence curves using 1 minus Kaplan-Meier estimates compared event rates across LA-related chamber metric categories: <25th percentile, 25th to 75th percentiles (reference), >75th percentile, and >95th percentile.

To illustrate the predictive importance of cardiac chamber volume indexes and chamber ratios alongside traditional cardiovascular risk factors for AF and stroke, we applied the least absolute shrinkage and selection operator (LASSO) regression. LASSO is a linear penalized regression technique that shrinks the coefficients of less informative or highly correlated variables toward zero, thereby performing automatic variable selection, reducing model complexity, and helping prevent overfitting when working with a large number of candidate predictors. This approach enables identification of the subset of predictors with the strongest independent prognostic association, improving interpretability in a clinical risk-prediction context. To further assess the robustness of variable selection in the presence of correlated predictors, we additionally implemented Elastic-Net regression, which combines L1 and L2 regularization penalties to enable variable selection while more effectively addressing multicollinearity among highly correlated chamber metrics.

To address multiple testing, we used a hierarchical approach. Based on prior validation and health care rationale, we prespecified continuous chamber metrics (LAVI, LA/RA ratio, LA/LV ratio) in pooled cohorts as primary analyses. Categorical thresholds, individual cohort analyses, and subgroup stratifications were designated secondary/exploratory. For sensitivity analysis, we applied Bonferroni correction ($\alpha=0.05/6=0.0083$) for 3 metrics tested across 2 outcomes (AF and stroke). All tests were 2-tailed.

RESULTS

Study Population

This study included 5670 participants from MESA and 1142 participants from FHS. The MESA cohort had a mean age of 62.1 ± 10.3 years, with 47.7% male participants, and demonstrated racial diversity: 38.9% White, 26.6% Black, 22.4% Hispanic, and 12.1% Chinese American. The FHS cohort was younger (mean age,

58.8 ± 8.4 years), with 44.4% male participants, and was predominantly White (99.8%). Table 1 presents comprehensive baseline characteristics stratified by the overall population and incident events during follow-up.

Incident Event Rates

Incident AF occurred in 1129 MESA participants (19.9%) across 77 925 person-years, yielding an incidence rate of 14 per 1000 person-years. In FHS, 173 participants (15.1%) had incident AF across 15 183 person-years (incidence rate: 11 per 1000 person-years). For stroke events, 314 MESA participants (5.5%) developed incident stroke across 84 309 person-years (incidence rate: 4 per 1000 person-years), while 51 FHS participants (4.5%) developed stroke across 15 816 person-years (incidence rate: 3 per 1000 person-years).

Cumulative Incidence Across Cardiac Chambers and Chamber Ratios Percentiles

Figures 2 and 3 illustrate cumulative incidence patterns for AF and stroke stratified by sex-specific percentiles of AI-derived cardiac parameters across pooled and individual cohorts. Participants in the highest percentile category (≥ 95 th) consistently demonstrated the highest cumulative incidence for both outcomes, followed by those in the ≥ 75 th percentile. FHS showed similar rank-order patterns to MESA. In pooled cohorts, cumulative AF incidence increased progressively across percentiles for all metrics, with LAVI showing the highest AF incidence (59%, 45.3 per 1000 person-years in the ≥ 95 th percentile). For stroke, the LA/RA ratio demonstrated the highest cumulative rates (17%, 10.1 per 1000 person-years in the ≥ 95 th percentile).

Risk Associations

Table 2 demonstrates associations between AI-measured cardiac chamber parameters and incident events across both cohorts. In pooled analysis, participants in the ≥ 95 th percentile showed significant adjusted hazard ratios for AF: 2.67 (95% CI, 2.23–3.20; $P<0.001$) for LAVI, 2.02 (95% CI, 1.68–2.42; $P<0.001$) for LA/LV ratio, and 1.83 (95% CI, 1.51–2.22; $P<0.001$) for LA/RA ratio. The associations for stroke were more modest but remained statistically significant, 1.98 (95% CI, 1.38–2.83; $P<0.001$) for LAVI, 1.83 (95% CI, 1.29–2.61; $P<0.001$) for LA/RA ratio, and 1.63 (95% CI, 1.13–2.33; $P<0.001$) for LA/LV ratio in the ≥ 95 th percentile. When examined as continuous variables, each 1-SD increment yielded adjusted hazard ratios for AF of 1.42 (95% CI, 1.35–1.50; $P<0.001$), 1.27 (95% CI, 1.21–1.34; $P<0.001$), and 1.22 (95% CI, 1.16–1.29; $P<0.001$) for LAVI, LA/LV ratio, and LA/RA ratio, respectively. For stroke, the corresponding adjusted hazard ratios were 1.30 (95% CI, 1.18–1.44; $P<0.001$),

Table 1. Baseline Characteristics of Participants in the MESA and the FHS, Overall and by Incident AF and Incident Stroke Over 17 Years

Variables	MESA			FHS		
	Overall (n=5670)	Incident AF (n=1129)	Incident stroke (n=314)	Overall (n=1142)	Incident AF (n=173)	Incident stroke (n=51)
Age, y	62.1±10.3	67.8±8.7	67.1±9.7	58.8±8.4	63.5±7.5	62.0±8.2
Sex, male	2706 (47.7%)	611 (54.1%)	158 (50.3%)	507 (44.4%)	98 (56.6%)	20 (39.2%)
Race and ethnicity						
White	2208 (38.9%)	521 (46.1%)	120 (38.2%)	988 (99.8%)	151 (100.0%)	41 (97.6%)
Chinese American	685 (12.1%)	141 (12.5%)	22 (7.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Black	1508 (26.6%)	255 (22.6%)	85 (27.1%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Hispanic	1269 (22.4%)	212 (18.8%)	87 (27.7%)	2 (0.2%)	0 (0.0%)	1 (2.4%)
Body mass index, kg/m ²	28.4±5.5	28.6±5.5	28.9±5.0	28.0±5.1	29.4±5.5	28.7±4.7
Waist circumference, cm	98.3±14.4	100.5±14.4	101.0±12.9	98.8±13.2	103.9±13.9	102.2±13.4
Creatinine, mg/dL	0.9 (0.8–1.1)	0.9 (0.8–1.1)	0.9 (0.8–1.1)	0.8 (0.7–1.0)	0.9 (0.8–1.0)	0.9 (0.8–1.0)
Diabetes	638 (11.3%)	160 (14.2%)	62 (19.7%)	98 (8.7%)	22 (12.8%)	1 (2.0%)
Current smokers	747 (13.2%)	122 (10.8%)	36 (11.5%)	101 (8.8%)	16 (9.2%)	6 (11.8%)
Blood pressure						
Systolic blood pressure, mm Hg	126±21	133±22	136±23	125±18	130±18	132±18
Diastolic blood pressure, mm Hg	72±10	72±10	74±10	74±9	73±9	74±8
Hypertension medication use	2094 (36.9%)	570 (50.5%)	158 (50.6%)	291 (25.5%)	63 (36.4%)	14 (27.5%)
Lipid profile						
Low-density lipoprotein, mg/dL	117±31	114±32	118±29	123±33	118±32	129±39
High-density lipoprotein, mg/dL	51±15	51±16	49±14	54±16	51±17	53±18
Triglycerides, mg/dL	113 (79–162)	113 (77–161)	123 (86–165)	114 (79–166)	119 (82–177)	137 (104–213)
Lipid-lowering medication Use	926 (16.4%)	239 (21.2%)	59 (19.0%)	195 (17.1%)	41 (23.7%)	10 (19.6%)
CAC scan-based AI chambers volumetry						
LA volume index, mL/m ²	30.3±8.5	34.1±10.5	33.7±10.7	35.2±9.6	42.3±11.7	40.2±12.3
LV volume index, mL/m ²	54.8±11.8	55.8±13.0	55.1±11.4	55.6±11.4	58.2±12.6	58.1±11.0
RA volume index, mL/m ²	34.5±8.9	36.4±10.5	35.1±9.2	37.0±10.3	41.9±12.5	37.7±11.2
RV volume index, mL/m ²	65.8±15.1	65.5±15.5	63.6±13.9	68.0±14.8	70.6±15.0	68.7±14.6
LV/RV ratio	0.8±0.1	0.9±0.1	0.9±0.1	0.8±0.1	0.8±0.1	0.9±0.2
LA/RA ratio	0.9±0.2	1.0±0.2	1.0±0.2	1.0±0.2	1.0±0.2	1.1±0.2
LA/LV ratio	0.6±0.2	0.6±0.2	0.6±0.2	0.6±0.1	0.7±0.2	0.7±0.2
Cardiovascular event risk						
CHARGE-AF	1.5 (0.6–3.8)	3.4 (1.6–6.5)	3.3 (1.3–6.7)	1.2 (0.6–2.5)	2.5 (1.3–4.9)	1.9 (1.0–3.7)
Framingham stroke risk profile	2.6 (0.9–6.7)	5.6 (2.6–9.9)	5.9 (2.3–11.5)	1.5 (0.8–3.8)	3.4 (1.4–7.0)	2.5 (1.2–5.0)

Data are presented as median (IQR) or mean±SD for continuous variables and frequency (percentage, %) for categorical variables. AF, atrial fibrillation; AI, artificial intelligence; CAC, coronary artery calcium; CHARGE-AF, Cohorts for Heart and Aging Research in Genomic Epidemiology Atrial Fibrillation risk model; FHS, Framingham Heart Study; LA, left atrium; LV, left ventricle; MESA, Multi-Ethnic Study of Atherosclerosis; RA, right atrium; RV, right ventricle.

1.28 (95% CI, 1.17–1.41; $P<0.001$), and 1.23 (95% CI, 1.11–1.37; $P<0.001$) for LAVI, LA/RA ratio, and LA/LV ratio, respectively. Associations were generally stronger in MESA compared with FHS, with FHS showing attenuated risks after multivariable adjustment, particularly for the LA/RA ratio in both AF and stroke models, and the LA/LV ratio in stroke prediction.

Subgroup Analysis

Tables S2 and S3 present detailed subgroup analyses. For AF prediction, most associations remained statistically

significant across demographic and clinical subgroups, including age categories, sex, smoking status, hypertension, diabetes, dyslipidemia, and racial and ethnic groups in adjusted models. Stroke subgroup analyses demonstrated more variable patterns with wider CIs due to fewer events.

Risk Reclassification

Table 3 and Table S4 summarize reclassification performance after incorporating AI-derived chamber measurements into established clinical risk models. For AF prediction using CHARGE-AF as the baseline, the

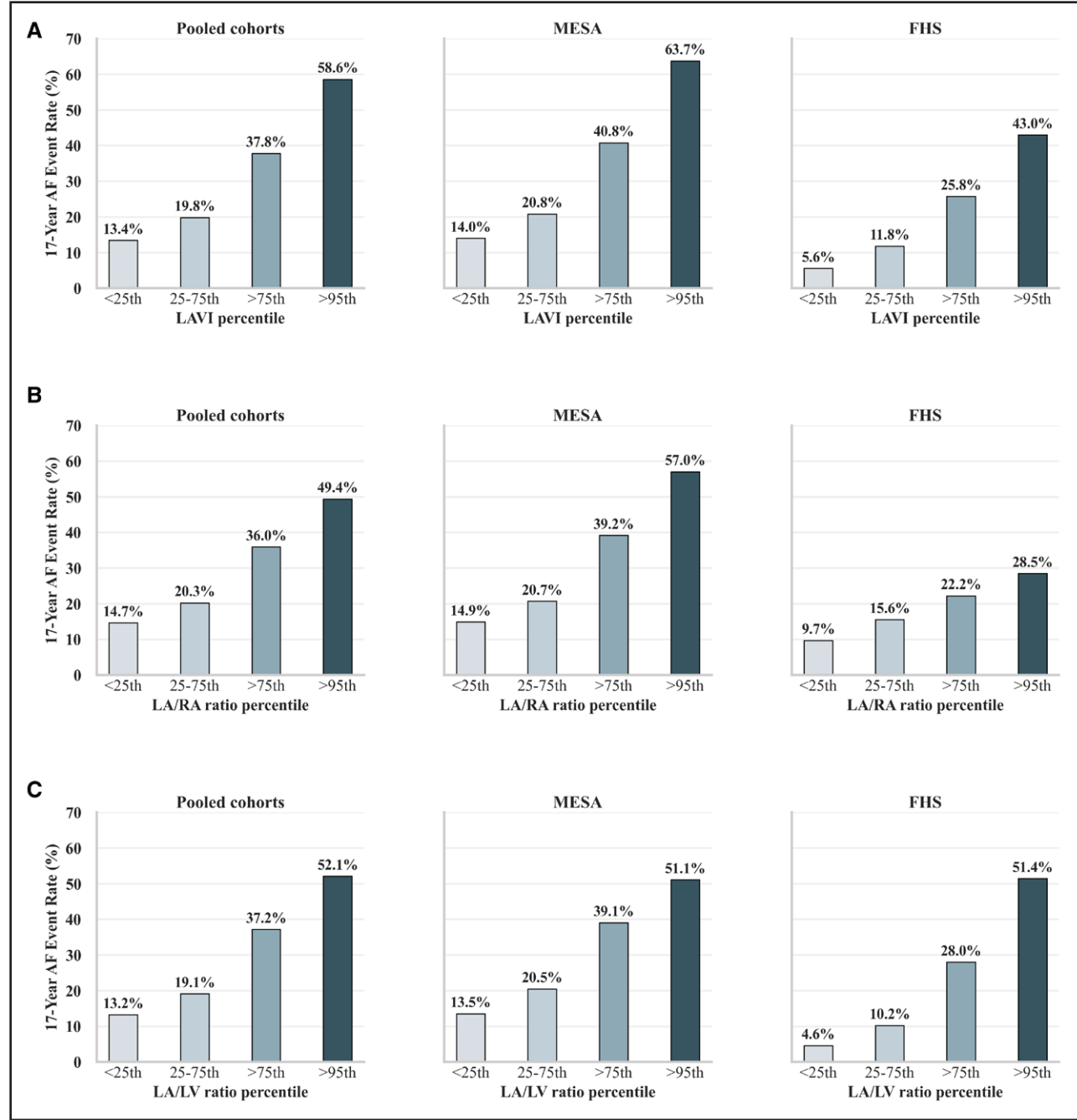


Figure 2. Seventeen-year cumulative atrial fibrillation (AF) rates by percentile for (A) left atrial (LA) volume index (LAVI), (B) LA/right atrial (RA) ratio, and (C) LA/left ventricle (LV) ratio in pooled MESA (Multi-Ethnic Study of Atherosclerosis)+FHS (Framingham Heart Study Offspring) cohorts (left), MESA (center), and FHS (right), showing increasing AF incidence at higher percentiles.

addition of continuous LAVI, LA/RA ratio, and LA/LV ratio significantly improved net reclassification in the pooled cohort, with NRIs of 0.48 (95% CI, 0.32–0.60; $P<0.001$) for LAVI, 0.37 (95% CI, 0.24–0.50; $P<0.001$) for LA/RA, and 0.40 (95% CI, 0.28–0.50; $P<0.001$) for LA/LV. These improvements were primarily driven by better classification of nonevents, with nonevent NRIs of 0.29 (95% CI, 0.24–0.36), 0.25 (95% CI, 0.19–0.30), and 0.24 (95% CI, 0.20–0.28) for LAVI, LA/RA, and LA/

LV, respectively. Event-related improvements were modest: 0.18 (95% CI, 0.03–0.28) for LAVI, 0.12 (95% CI, 0.00–0.23) for LA/RA, and 0.16 (95% CI, 0.05–0.26) for LA/LV. All 3 continuous measures demonstrated significant improvement in MESA, while in FHS, LAVI, and LA/LV showed significant reclassification improvement, but the LA/RA ratio did not reach statistical significance.

For stroke prediction using the FSRP, continuous LAVI, LA/RA ratio, and LA/LV ratio also showed

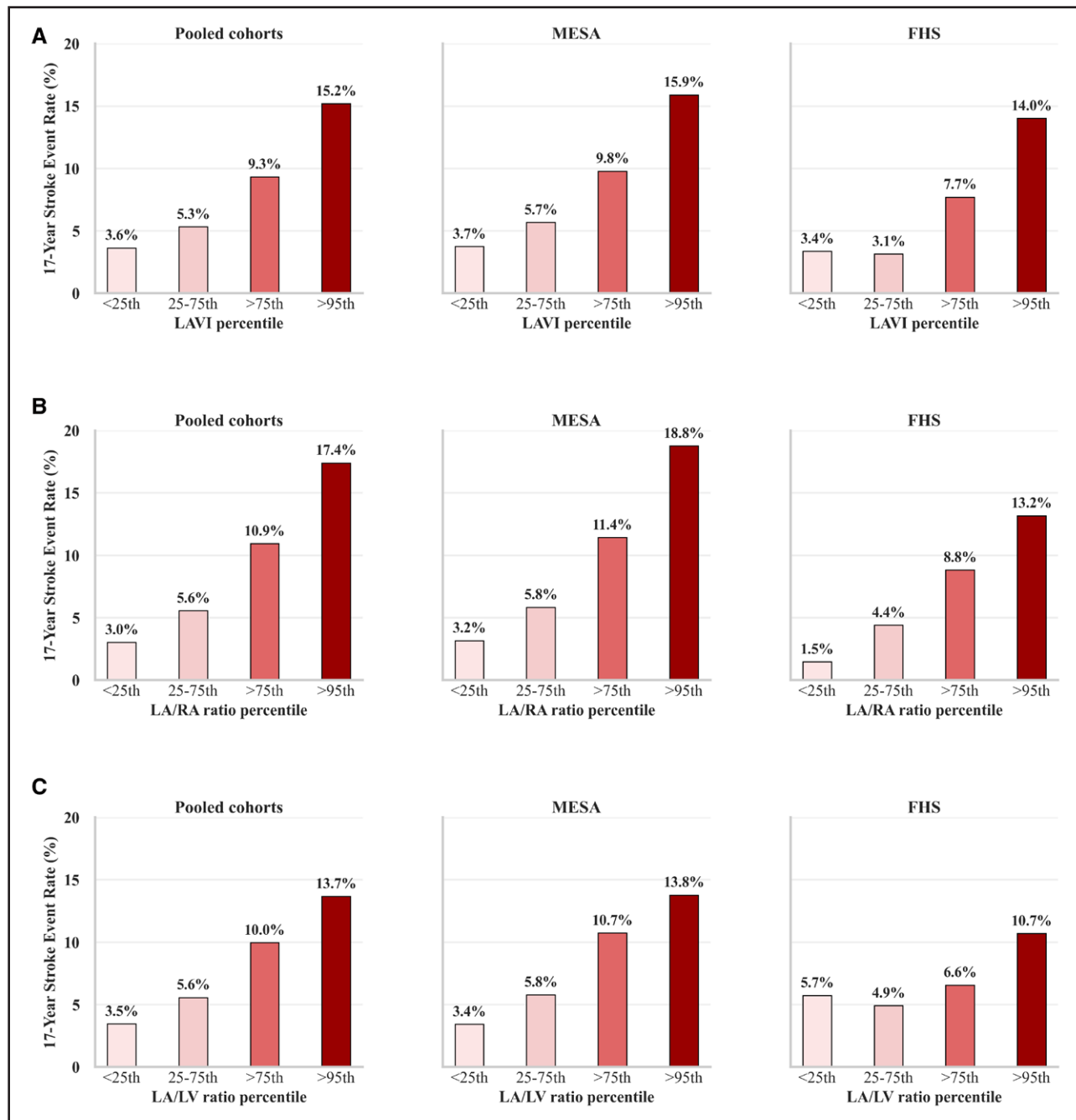


Figure 3. Seventeen-year cumulative stroke rates by percentile for (A) left atrial (LA) volume index (LAVI), (B) LA/right atrial (RA) ratio, and (C) LA/left ventricle (LV) ratio in pooled MESA (Multi-Ethnic Study of Atherosclerosis)+FHS (Framingham Heart Study Offspring) cohorts (left), MESA (center), and FHS (right), showing increasing stroke incidence at higher percentiles.

reclassification improvement in the pooled cohort: 0.30 (95% CI, 0.14–0.45; $P < 0.001$) for LAVI, 0.39 (95% CI, 0.25–0.53; $P < 0.001$) for LA/RA, and 0.21 (95% CI, 0.07–0.38; $P < 0.001$) for LA/LV. These gains were primarily attributable to nonevent reclassification, with nonevent NRIs of 0.24 (95% CI, 0.18–0.29), 0.28 (95% CI, 0.24–0.33), and 0.23 (95% CI, 0.18–0.29), respectively. Event reclassification did not contribute significantly, with event NRIs of 0.07 (95% CI, –0.08 to 0.19) for LAVI, 0.11 (95% CI, –0.02 to 0.23) for

LA/RA, and –0.02 (95% CI, –0.13 to 0.11) for LA/LV. Within individual cohorts, LAVI and LA/RA ratio provided significant benefit in MESA (NRIs, 0.28 and 0.37, respectively), while in FHS, LAVI showed a borderline significant effect.

Incremental Discrimination

Table 4 presents time-dependent AUC values for risk prediction models with and without cardiac chamber

Table 2. Associations of CAC Scan-Based, AI-Measured LA-Related Chamber Metrics and Ratios With Incident Atrial Fibrillation and Stroke in MESA, FHS, and Pooled MESA+FHS Cohorts

	Atrial Fibrillation				Stroke			
	Index			HR per 1-SD increment	Index			HR per 1-SD increment
	25th–75th	≥75th	≥95th		25th–75th	≥75th	≥95th	
LA volume index								
MESA								
Incidence rate (per 100 PYs)	1.3	2.8	5.1	...	0.3	0.6	1.1	...
Unadjusted HR (95% CI)	Ref	2.33 (2.05–2.65)	4.63 (3.81–5.62)	1.72 (1.63–1.80)	Ref	1.88 (1.47–2.41)	3.29 (2.24–4.82)	1.57 (1.43–1.72)
Model 1 HR (95% CI)	Ref	1.62 (1.42–1.85)	2.58 (2.10–3.17)	1.37 (1.30–1.45)	Ref	1.31 (1.01–1.69)	1.88 (1.25–2.82)	1.28 (1.15–1.43)
FHS								
Incidence rate (per 100 PYs)	0.8	1.9	3.5	...	0.2	0.5	0.8	...
Unadjusted HR (95% CI)	Ref	2.50 (1.81–3.46)	4.87 (3.32–7.15)	2.02 (1.80–2.26)	Ref	2.42 (1.32–4.46)	4.23 (2.04–8.77)	1.59 (1.28–1.99)
Model 1 HR (95% CI)	Ref	1.90 (1.35–2.67)	3.21 (2.12–4.89)	1.70 (1.48–1.94)	Ref	2.31 (1.19–4.49)	3.37 (1.48–7.67)	1.46 (1.13–1.89)
Pooled cohort								
Incidence rate (per 100 PYs)	1.2	2.5	4.5	...	0.3	0.6	1.0	...
Unadjusted HR (95% CI)	Ref	2.25 (2.00–2.54)	4.33 (3.66–5.13)	1.72 (1.65–1.80)	Ref	1.88 (1.50–2.35)	3.21 (2.31–4.46)	1.57 (1.43–1.72)
Model 1 HR (95% CI)	Ref	1.67 (1.47–1.88)	2.66 (2.23–3.17)	1.42 (1.34–1.49)	Ref	1.39 (1.10–1.76)	1.96 (1.38–2.77)	1.28 (1.15–1.43)
Model 2 HR (95% CI)	Ref	1.66 (1.46–1.88)	2.67 (2.23–3.20)	1.42 (1.35–1.50)	Ref	1.39 (1.10–1.77)	1.98 (1.38–2.83)	1.30 (1.18–1.44)
LA/RA ratio								
MESA								
Incidence rate (per 100 PYs)	1.3	2.6	4.4	...	0.3	0.7	1.2	...
Unadjusted HR (95% CI)	Ref	2.22 (1.95–2.52)	3.97 (3.25–4.86)	1.51 (1.44–1.58)	Ref	2.01 (1.58–2.55)	3.52 (2.44–5.07)	1.51 (1.40–1.64)
Model 1 HR (95% CI)	Ref	1.39 (1.22–1.59)	1.99 (1.61–2.47)	1.23 (1.16–1.30)	Ref	1.32 (1.02–1.69)	1.94 (1.32–2.86)	1.27 (1.15–1.41)
FHS								
Incidence rate (per 100 PYs)	1.0	1.5	2.2	...	0.2	0.5	0.7	...
Unadjusted HR (95% CI)	Ref	1.59 (1.16–2.18)	2.35 (1.55–3.58)	1.38 (1.21–1.57)	Ref	2.19 (1.22–3.91)	3.05 (1.44–6.45)	1.54 (1.23–1.91)
Model 1 HR (95% CI)	Ref	1.05 (0.75–1.47)	1.27 (0.80–1.99)	1.15 (0.99–1.34)	Ref	1.77 (0.96–3.29)	1.78 (0.76–4.18)	1.37 (1.07–1.75)
Pooled cohort								
Incidence rate (per 100 PYs)	1.2	2.3	3.6	...	0.3	0.6	1.0	...
Unadjusted HR (95% CI)	Ref	2.05 (1.82–2.31)	3.37 (2.81–4.03)	1.47 (1.41–1.54)	Ref	1.98 (1.59–2.47)	3.20 (2.32–4.43)	1.51 (1.40–1.64)
Model 1 HR (95% CI)	Ref	1.36 (1.20–1.53)	1.87 (1.55–2.26)	1.23 (1.17–1.30)	Ref	1.35 (1.07–1.70)	1.83 (1.29–2.59)	1.27 (1.15–1.41)
Model 2 HR (95% CI)	Ref	1.34 (1.18–1.51)	1.83 (1.51–2.22)	1.22 (1.16–1.29)	Ref	1.35 (1.07–1.70)	1.83 (1.29–2.61)	1.28 (1.17–1.41)
LA/LV ratio								
MESA								
Incidence rate (per 100 PYs)	1.3	2.6	3.7	...	0.3	0.6	0.8	...
Unadjusted HR (95% CI)	Ref	2.22 (1.95–2.52)	3.26 (2.68–3.97)	1.48 (1.42–1.55)	Ref	1.81 (1.42–2.30)	2.48 (1.70–3.63)	1.46 (1.34–1.60)
Model 1 HR (95% CI)	Ref	1.43 (1.25–1.64)	1.75 (1.43–2.15)	1.21 (1.14–1.28)	Ref	1.35 (1.05–1.74)	1.62 (1.09–2.40)	1.24 (1.11–1.39)
FHS								
Incidence rate (per 100 PYs)	0.6	2.0	4.5	...	0.3	0.4	0.7	...
Unadjusted HR (95% CI)	Ref	3.24 (2.34–4.50)	7.81 (5.22–11.67)	1.89 (1.69–2.11)	Ref	1.59 (0.91–2.79)	2.98 (1.37–6.48)	1.48 (1.17–1.85)
Model 1 HR (95% CI)	Ref	2.44 (1.70–3.50)	4.72 (2.97–7.50)	1.94 (1.68–2.23)	Ref	1.23 (0.66–2.32)	1.91 (0.80–4.57)	1.25 (0.94–1.66)
Pooled cohort								
Incidence rate (per 100 PYs)	1.1	2.5	3.9	...	0.3	0.6	0.8	...
Unadjusted HR (95% CI)	Ref	2.28 (2.03–2.57)	3.77 (3.17–4.48)	1.52 (1.46–1.59)	Ref	1.74 (1.39–2.16)	2.54 (1.81–3.56)	1.46 (1.34–1.60)
Model 1 HR (95% CI)	Ref	1.53 (1.36–1.73)	2.04 (1.71–2.45)	1.28 (1.21–1.35)	Ref	1.30 (1.03–1.64)	1.64 (1.15–2.33)	1.24 (1.11–1.39)
Model 2 HR (95% CI)	Ref	1.52 (1.34–1.72)	2.02 (1.68–2.42)	1.27 (1.21–1.34)	Ref	1.30 (1.03–1.64)	1.63 (1.14–2.33)	1.23 (1.11–1.37)

Model 1: adjusted for the following covariates—age, sex, race and ethnicity, systolic and diastolic blood pressure, antihypertensive medication use, LDL cholesterol, HDL cholesterol, triglycerides, lipid-lowering medication use, diabetes, and current smoking status. Model 2: includes all covariates in model 1, with additional adjustment for the study cohort. AI, artificial intelligence; CAC, coronary artery calcium; FHS, Framingham Heart Study; HR, hazard ratio; LA/LV, left atrium–left ventricle ratio; LA/RA, left atrium–right atrium ratio; MESA, Multi-Ethnic Study of Atherosclerosis; PY, person-years.

Table 3. Model Performance After Adding CAC scan-based, AI-measured LA-related chamber metrics and ratios to CHARGE-AF for AF Prediction and to Framingham Stroke Risk Profile for Stroke Prediction: NRI in MESA, FHS, and Pooled MESA+FHS Cohorts

	Pooled cohorts	MESA	FHS
	Overall	Overall	Overall
AF			
LAVI added to CHARGE-AF			
Overall	0.48 (0.32–0.60)	0.43 (0.28–0.58)	0.84 (0.58–1.14)
≥75th	0.56 (0.44–0.67)	0.53 (0.39–0.67)	0.85 (0.62–1.07)
≥95th	0.41 (0.30–0.51)	0.35 (0.25–0.45)	0.77 (0.43–1.10)
LA/RA ratio added to CHARGE-AF			
Overall	0.37 (0.24–0.50)	0.42 (0.27–0.53)	0.21 (–0.13 to 0.55)
≥75th	0.46 (0.34–0.57)	0.49 (0.36–0.62)	0.31 (–0.24 to 0.64)
≥95th	0.27 (0.17–0.37)	0.26 (0.16–0.36)	0.32 (–0.14 to 0.59)
LA/LV ratio added to CHARGE-AF			
Overall	0.40 (0.28–0.50)	0.34 (0.18–0.49)	0.63 (0.34–0.99)
≥75th	0.52 (0.41–0.66)	0.45 (0.30–0.57)	0.99 (0.80–1.18)
≥95th	0.29 (–0.12 to 0.39)	0.22 (–0.54 to 0.32)	0.67 (0.36–0.95)
Stroke			
LAVI added to FSRP			
Overall	0.30 (0.14–0.45)	0.28 (0.11–0.44)	0.30 (0.00–0.63)
≥75th	0.46 (0.33–0.57)	0.47 (0.33–0.63)	0.37 (0.04–0.72)
≥95th	0.22 (0.13–0.32)	0.20 (–0.45 to 0.31)	0.33 (–0.01 to 0.64)
LA/RA ratio added to FSRP			
Overall	0.39 (0.25–0.53)	0.37 (0.18–0.55)	0.32 (–0.04 to 0.69)
≥75th	0.45 (0.30–0.58)	0.48 (0.32–0.64)	0.33 (–0.01 to 0.73)
≥95th	0.23 (0.13–0.36)	0.24 (–0.04–0.36)	0.17 (–0.06 to 0.41)
LA/LV ratio added to FSRP			
Overall	0.21 (0.07–0.38)	0.19 (0.04–0.40)	0.20 (–0.10 to 0.57)
≥75th	0.29 (0.14–0.42)	0.30 (0.16–0.44)	0.21 (–0.09 to 0.56)
≥95th	0.19 (0.09–0.30)	0.17 (–0.12 to 0.27)	0.28 (–0.04 to 0.65)

AF, atrial fibrillation; AI, artificial intelligence; CAC, Cohorts for Heart and Aging Research in Genomic Epidemiology Atrial Fibrillation risk model; FHS, coronary artery calcium; CHARGE-AF, Framingham Heart Study; FSRP, Framingham Stroke Risk Profile; LA, left atrial volume index; LV, left atrial/LV left ventricular; MESA, Multi-Ethnic Study of Atherosclerosis; NRI, net reclassification improvement; RA, right atrial.

measurements. For AF prediction at 5 years, CHARGE-AF alone achieved AUCs of 0.760 (95% CI, 0.735–0.786) in pooled cohorts, 0.766 (95% CI, 0.739–0.793) in MESA, and 0.718 (95% CI, 0.642–0.794) in FHS. In FHS, continuous LAVI significantly improved discrimination to an AUC of 0.814 (95% CI, 0.751–0.877; $P=0.008$) and continuous LA/LV ratio to 0.796 (95% CI, 0.728–0.865; $P=0.035$). In the pooled cohort and MESA, the addition of chamber measurements resulted in modest improvements in discrimination that did not reach statistical significance.

For stroke prediction at 10 years, the FSRP alone demonstrated AUCs of 0.735 (95% CI, 0.701–0.770) in pooled cohorts, 0.757 (95% CI, 0.720–0.793) in MESA, and 0.641 (95% CI, 0.558–0.725) in FHS. The addition of chamber measurements resulted in modest improvements in discrimination that did not reach statistical significance in any cohort.

Model Calibration

Table S5 presents calibration performance for clinical risk models with and without AI-derived chamber measurements. For AF prediction, CHARGE-AF alone demonstrated adequate baseline calibration in the pooled cohort (slope, 1.059; Brier score, 0.0189) and MESA (slope, 1.027; Brier score, 0.0192). Adding chamber metrics resulted in only minimal calibration improvements, with continuous LAVI achieving near-perfect calibration in the pooled cohort (slope, 0.997) and dichotomized LAVI at ≥75th and ≥95th percentiles in MESA (slopes, 0.999 and 1.002). Brier scores remained largely unchanged across models.

In contrast, for stroke prediction, FSRP alone exhibited poor baseline calibration across all cohorts with slopes substantially below 1.0 (pooled, 0.448; MESA, 0.469; FHS, 0.240), indicating systematic overestimation of

Table 4. Time-Dependent AUC for CHARGE-AF and FSRP Risk Models, Alone and With Added CAC Scan-Based, AI-Measured LA-Related Chamber Metrics and Ratios (LAVI, LA/RA Ratio, LA/LV Ratio—as Continuous Values and at ≥ 75 th and ≥ 95 th Percentiles), for Prediction of Atrial Fibrillation at 5 Years (CHARGE-AF) and Stroke at 10 Years (FSRP) in the MESA, FHS, and Pooled MESA+FHS Cohorts

	Pooled cohort		MESA		FHS	
	AUC (95% CI)	P value	AUC (95% CI)	P value	AUC (95% CI)	P value
AF						
CHARGE-AF alone	0.760 (0.735–0.786)	NA	0.766 (0.739–0.793)	NA	0.718 (0.642–0.794)	NA
LAVI added to CHARGE-AF						
Overall	0.761 (0.732–0.790)	0.964	0.754 (0.723–0.786)	0.457	0.814 (0.751–0.877)	0.008
≥ 75 th	0.767 (0.742–0.793)	0.580	0.767 (0.739–0.795)	0.938	0.788 (0.725–0.852)	0.052
≥ 95 th	0.777 (0.751–0.803)	0.210	0.778 (0.751–0.805)	0.370	0.757 (0.676–0.839)	0.333
LA/RA ratio added to CHARGE-AF						
Overall	0.756 (0.728–0.784)	0.764	0.764 (0.734–0.793)	0.887	0.711 (0.629–0.794)	0.857
≥ 75 th	0.759 (0.733–0.784)	0.907	0.766 (0.739–0.793)	0.992	0.719 (0.643–0.795)	0.982
≥ 95 th	0.762 (0.736–0.788)	0.881	0.770 (0.743–0.797)	0.779	0.717 (0.639–0.795)	0.977
LA/LV ratio added to CHARGE-AF						
Overall	0.756 (0.729–0.784)	0.788	0.756 (0.726–0.785)	0.488	0.796 (0.728–0.865)	0.035
≥ 75 th	0.756 (0.730–0.783)	0.777	0.760 (0.732–0.787)	0.675	0.754 (0.676–0.833)	0.359
≥ 95 th	0.766 (0.741–0.791)	0.650	0.769 (0.743–0.795)	0.810	0.739 (0.658–0.819)	0.613
Stroke						
FSRP alone	0.735 (0.701–0.770)	NA	0.757 (0.720–0.793)	NA	0.641 (0.558–0.725)	NA
LAVI added to FSRP						
Overall	0.746 (0.712–0.780)	0.522	0.761 (0.725–0.798)	0.798	0.656 (0.562–0.750)	0.745
≥ 75 th	0.752 (0.719–0.784)	0.451	0.771 (0.738–0.805)	0.419	0.665 (0.574–0.756)	0.593
≥ 95 th	0.748 (0.715–0.782)	0.334	0.766 (0.731–0.801)	0.614	0.661 (0.573–0.750)	0.646
LA/RA ratio added to FSRP						
Overall	0.752 (0.719–0.786)	0.318	0.767 (0.731–0.804)	0.576	0.674 (0.594–0.754)	0.427
≥ 75 th	0.749 (0.717–0.782)	0.408	0.768 (0.734–0.802)	0.527	0.665 (0.576–0.753)	0.594
≥ 95 th	0.744 (0.710–0.778)	0.610	0.764 (0.728–0.800)	0.694	0.656 (0.572–0.740)	0.733
LA/LV ratio added to FSRP						
Overall	0.736 (0.702–0.770)	0.967	0.754 (0.718–0.790)	0.897	0.629 (0.533–0.726)	0.799
≥ 75 th	0.734 (0.700–0.768)	0.941	0.752 (0.716–0.788)	0.811	0.644 (0.555–0.733)	0.951
≥ 95 th	0.739 (0.705–0.773)	0.813	0.761 (0.725–0.797)	0.804	0.639 (0.552–0.726)	0.963

Time-dependent AUCs are compared against the base model (CHARGE-AF alone for AF; FSRP alone for stroke). P values are from the DeLong test comparing the AUC of each augmented model to that of the base model CHARGE-AF alone for AF predictions; FSRP alone for stroke predictions. AF, atrial fibrillation; AI, artificial intelligence; AUC, Cohorts for Heart and Aging Research in Genomic Epidemiology Atrial Fibrillation risk model; FHS, coronary artery calcium; CHARGE-AF, Framingham Heart Study; FSRP, Framingham Stroke Risk Profile; LA, left atrial volume index; LAVI, left ventricular; MESA, Multi-Ethnic Study of Atherosclerosis; RA, under the curve; CAC, right atrial.

stroke risk. Adding chamber measurements substantially improved calibration performance. In the pooled cohort, continuous chamber metrics improved slopes to 0.893 (LAVI), 0.834 (LA/RA ratio), and 0.902 (LA/LV ratio), with reduced Brier scores (0.0155–0.0156 versus 0.0167). In MESA, the LA/LV ratio at the ≥ 95 th percentile achieved near-perfect calibration (slope, 0.951). In FHS, continuous LAVI markedly improved calibration from 0.240 to 1.130.

Feature Selection Analysis

Figure 4 displays LASSO regression coefficients for AF and stroke prediction across MESA, FHS, and pooled cohorts. For AF prediction, age consistently emerged

as the strongest predictor across all cohorts, followed by LAVI, which was selected as an important feature in all 3 analyses. Additional variables, including blood pressure, diabetes, and other chamber ratios, showed varying importance across cohorts. For stroke prediction, systolic blood pressure and age dominated as predictors, with the LA/RA ratio consistently selected across all cohorts as an important feature. This variable selection pattern demonstrates that cardiac chamber measurements, particularly LAVI and LA/RA ratio, provide additional predictive value beyond traditional cardiovascular risk factors for both clinical outcomes. Elastic-Net regression produced coefficient patterns closely matching those obtained using LASSO (Figure S7).

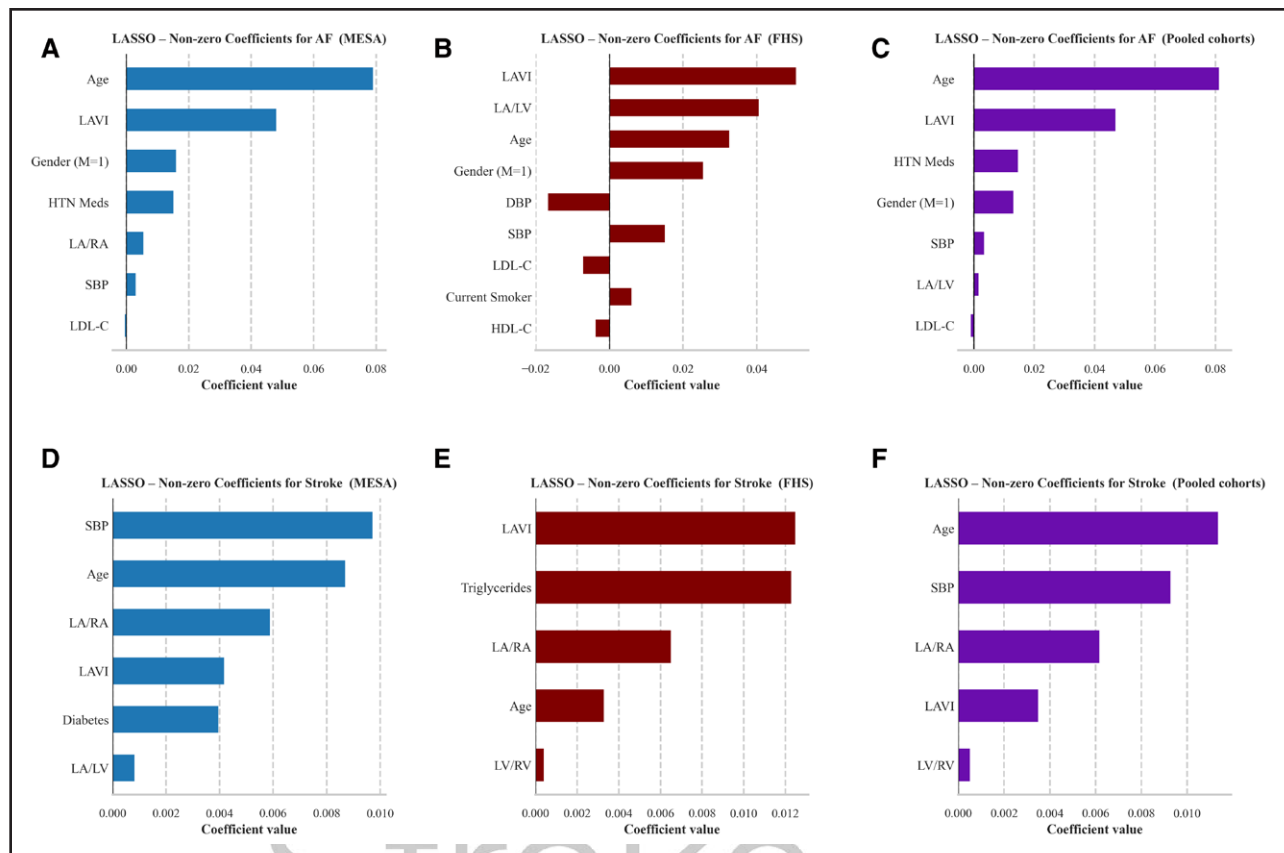


Figure 4. Least absolute shrinkage and selection operator (LASSO) regression coefficients for atrial fibrillation (AF) and stroke models.

A through **C**, Nonzero coefficients selected by LASSO regression for AF prediction in **(A)** the MESA (Multi-Ethnic Study of Atherosclerosis) cohort, **(B)** the FHS (Framingham Heart Study) cohort, and **(C)** both cohorts pooled. **D** through **F**, Same for the stroke prediction model: **(D)** MESA, **(E)** FHS, and **(F)** pooled cohorts. DBP indicates diastolic blood pressure; HDL-C, high-density lipoprotein cholesterol; LA, left atrial; LAVI, left atrial volume index; LDL-C, low-density lipoprotein cholesterol; LV, left ventricle; RA, right atrial; and SBP, systolic blood pressure.

DISCUSSION

To the best of our knowledge, this study is the first to assess the role of AI-derived LA-related measurements and ratios from CAC scans in 2 large, well-characterized cohorts. This study demonstrated that AI-derived LA-related cardiac chamber measurements opportunistically measured from CAC scans provided incremental value for predicting incident AF and stroke over long-term follow-up. Our findings reveal that LAVI, LA/RA ratio, and LA/LV ratio are independent predictors of both outcomes and demonstrated consistent predictive value across diverse demographic and clinical subgroups. These AI-derived measurements enhanced risk reclassification beyond established prediction models (CHARGE-AF and FSRP). Although discrimination improvements were modest and nonsignificant, the addition of chamber measurements substantially improved model calibration for stroke prediction, whereas AF prediction models maintained adequate calibration both with and without added metrics. Feature importance analysis using LASSO regression confirmed LAVI as the

most consistent chamber measurement predictor for AF across all cohorts, while LA/RA ratio emerged as a key predictor for stroke risk.

The present study builds on our previous work demonstrating that AI-derived LA volumetry from CAC scans performs comparably to expert CMR measurements for AF and stroke prediction.^{10,11} These findings are particularly significant given the widespread availability of CAC scanning and the potential for opportunistic cardiovascular risk assessment without additional radiation exposure or cost.

Our findings confirm that LA volumetry measured via AI from CAC scans demonstrates comparable predictive value to the well-established associations previously demonstrated using echocardiography and cardiovascular magnetic resonance imaging. The Copenhagen Heart Study, which followed over 1000 participants for 16 years, demonstrated that LA volumetric and mechanical coupling independently predicted incident AF, even after comprehensive multivariable adjustment using echocardiography.²⁶ Similarly, a meta-analysis on AF recurrence following direct current cardioversion

revealed that larger LAVI measured echocardiographically significantly predicted AF recurrence.²⁷ In terms of stroke prediction, a large-scale echocardiographic study has consistently shown that LAVI values ≥ 32 mL/m² independently predict ischemic stroke in elderly populations.²⁸ A comprehensive meta-analysis involving 66 007 participants found that LA enlargement was associated with a 68% increased risk of stroke. Furthermore, each 1 cm increase in LA diameter was linked to a 24% increase in stroke odds.⁵

Our investigation of LA/RA and LA/LV ratios provides novel insights into chamber coupling relationships that may reflect distinct pathophysiological processes not captured by isolated chamber measurements. The LASSO and Elastic-Net regression analysis for feature importance in our study demonstrated that the LA/RA ratio emerged as a key predictor for stroke risk and was superior to isolated LAVI, suggesting that inter-atrial relationships may capture unique aspects of atrial cardiomyopathy and bi-atrial remodeling patterns not fully reflected by single-chamber measurements. The concept of atrioventricular coupling has gained increasing recognition in cardiovascular disease prediction, with previous studies from MESA demonstrating that the left atrioventricular coupling index (defined as the LA/LV ratio) independently predicts incident heart failure in asymptomatic participants and improves risk discrimination.^{14,29} Although the LA/LV ratio for heart failure prediction has been established, the LA/RA ratio as a predictor of AF represents a recent novel finding. A recent study by Hoballah et al¹⁶ in the MESA cohort demonstrated that the LA/RA ratio captured from CMR independently predicted incident AF and heart failure in individuals without baseline cardiovascular disease. Our findings extend this emerging concept by demonstrating that LA/RA and LA/LV ratios derived opportunistically from CAC scans using AI maintain their predictive value in the context of new onset AF and stroke prediction, with significant associations that persist after comprehensive risk adjustment.

Our findings demonstrate that AI-derived chamber measurements provide meaningful improvements to established risk prediction models, primarily through enhanced risk reclassification rather than discrimination. This pattern is expected when adding biomarkers to well-validated scores that already achieve good baseline performance (AUCs ≈ 0.72 – 0.76).

The CHARGE-AF score, developed from pooled data from 3 large American cohorts (ARIC, CHS, and FHS), including 18 556 participants, achieved good discrimination for AF prediction.²⁴ The addition of LAVI, LA/RA ratio, and LA/LV ratio to CHARGE-AF resulted in significant net reclassification improvement, with the most substantial benefits observed for accurate down-classification of nonevents. Importantly, the predominant contribution of nonevent reclassification has practical clinical value

beyond model performance metrics. Accurate identification of truly low-risk individuals could reduce unnecessary surveillance in primary prevention populations, allowing more efficient resource allocation and targeted monitoring of those at genuinely elevated risk. Although discrimination improvements (AUC) in the pooled cohort were modest and nonsignificant, continuous LAVI and LA/LV ratio significantly improved discrimination in FHS, suggesting potential variability in predictive utility across populations with different baseline risk profiles.

For stroke prediction, the FSRP has been extensively validated across multiple populations, though with variable performance depending on population characteristics.²⁵ Our findings demonstrate that the LA/RA ratio provides the most consistent improvements in terms of enhanced reclassification for stroke prediction.

CHARGE-AF demonstrated adequate baseline calibration that remained largely unchanged with chamber metrics added. In contrast, the FSRP exhibited poor baseline calibration with systematic risk overestimation, which was substantially improved by incorporating chamber measurements.

The clinical importance of AI-derived LA-related measurements from CAC scans offers potential for cardiovascular disease prevention through opportunistic screening. Integrating automated chamber analysis provides comprehensive risk evaluation without additional cost, radiation, or protocol modifications. Critically, chamber volume ratios such as LA/RA and LA/LV ratios are independent of body surface area adjustment, addressing a key limitation where BSA data are often unavailable in Picture Archiving and Communication Systems systems. This enables immediate clinical implementation, as ratios can be automatically calculated during routine CAC scoring to flag high-risk patients for targeted interventions. Our subgroup analyses confirm predictive value across diverse populations, supporting broad generalizability. The ability to identify asymptomatic individuals at elevated AF and stroke risk through opportunistic screening facilitates earlier preventive strategies, including enhanced surveillance, lifestyle modifications, and anticoagulation considerations. The automated AI analysis enables scalable implementation without additional radiologist burden, potentially capturing high-risk individuals who might otherwise not undergo formal cardiovascular assessment, thereby improving population-level prevention strategies and addressing healthcare disparities.

Limitations

Several important limitations warrant acknowledgment. First, the observational design precludes causal inferences regarding the relationship between AI-derived chamber measurements and cardiovascular outcomes, and the temporal relationship between chamber

enlargement and outcome development requires further investigation. Our measurements represent single time points, and the dynamic nature of cardiac remodeling suggests that serial assessments might provide enhanced predictive value for risk stratification. Second, there are technical considerations regarding imaging protocols. In MESA, baseline CT imaging was performed between 2000 and 2002 using EBCT or earlier-generation MDCT systems, whereas contemporary CAC assessments typically use more advanced MDCT scanners. We observed differences in AI-measured cardiac chamber volumetry when measured using different ECG gating methods (RR-interval) used in MESA for MDCT (50%) and EBCT (80%). This discrepancy resulted in significant differences in LA volume between participants scanned with EBCT versus MDCT (57.4 cm³ versus 65.4 cm³, respectively; $P<0.0001$). Similarly, RA volumes differed significantly between EBCT versus MDCT (66.9 cm³ versus 75.0 cm³, respectively; $P<0.0001$). In contrast, LV volume measurements were less affected by scanner type ($P>0.05$). Although interaction terms between LA volume and scanner type were tested and found to be nonsignificant for outcome prediction, the extent of the impact of scanner type on our findings remains uncertain and warrants further investigation. Third, in FHS, CAC scans were available for only a subset of the cohort, which could introduce selection bias and limit the generalizability of our FHS-specific findings. Fourth, while CAC scanning is increasingly used, it is not routinely available to many individuals in the United States without health insurance coverage. In contrast, over 19 million nongated, noncontrast chest CT scans are performed annually, often for lung cancer screening or other thoracic indications.³⁰ Although these scans represent a potentially valuable source for opportunistic cardiac assessment, they pose specific challenges: the cardiac phase at the time of imaging is typically unknown, and the prolonged acquisition duration can lead to variability across reconstructed slabs. This variability may affect volumetric accuracy and should be addressed in future technical validation studies.

Fifth, our stroke outcome combined ischemic and hemorrhagic stroke as a composite end point. Ischemic and hemorrhagic strokes have distinct pathophysiological mechanisms and risk factors, which may have attenuated the true association between atrial chamber enlargement and cardioembolic events. Finally, outcome ascertainment limitations may have affected our results, particularly in MESA, where AF identification relied on ICD codes. ICD-based AF diagnosis has known limitations with positive predictive values of 70% to 96% and a median sensitivity of 79%, likely resulting in missed AF cases.³¹ Most importantly, ICD-10 coding does not capture undiagnosed AF or recently diagnosed cases not yet coded, potentially missing a significant number of individuals with paroxysmal AF, which could have

underestimated the true association between chamber measurements and AF risk.

Future Directions

Future research should prioritize several key areas to advance clinical implementation of AI-derived cardiac chamber measurements. Prospective validation studies in diverse populations are essential to confirm generalizability across different ethnic groups and healthcare systems, while prospective intervention trials are critically needed to demonstrate that knowledge of elevated chamber measurements translates into improved clinical outcomes through targeted preventive strategies. Cost-effectiveness analyses will be crucial for supporting healthcare system adoption of AI-enabled opportunistic screening programs. The development of integrated risk prediction models incorporating multiple AI-derived biomarkers from CAC scans, including coronary calcification patterns and comprehensive chamber analysis, may provide superior risk stratification compared with individual measurements alone. Additionally, research should focus on optimizing clinical decision-making algorithms that seamlessly integrate AI-derived measurements with traditional risk factors for personalized cardiovascular assessment. Finally, longitudinal studies examining serial chamber measurements could elucidate cardiac remodeling dynamics and enhance predictive accuracy, while technical validation across different scanner manufacturers will be crucial for widespread clinical implementation.

Conclusions

In conclusion, our study demonstrates the incremental predictive value of AI-derived LA volumetric metrics and chamber ratios (LA/RA and LA/LV) from routine CAC scans for long-term prediction of incident AF and stroke. By leveraging automated AI-driven chamber analysis in 2 large, diverse, and well-characterized cohorts (MESA and FHS), we validated these metrics as robust independent predictors, showing meaningful enhancements in risk stratification beyond established clinical risk scores (CHARGE-AF and FSRP). Importantly, the ability to opportunistically identify high-risk individuals without additional imaging or radiation exposure underscores the transformative clinical potential of integrating AI-enabled chamber volumetry into routine cardiovascular screening practices, ultimately informing targeted preventive strategies and potentially reducing global burdens associated with AF and stroke.

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Supplemental Material

About AI-CVD: A Comprehensive CVD Risk Assessment based on Artificial Intelligence-Enabled Imaging Biomarkers from Coronary Artery Calcium Scans
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